Nils Matteson

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Systems engineer focused on GPU inference and ML infrastructure. Creator of **thaw**, open-source LLM-inference infrastructure (16 PyPI releases, upstream vLLM PR #44074, sole-author **arXiv** paper). Seeking a **Summer 2027 SWE/MLE internship** in GPU inference, distributed systems, or ML infrastructure.

Education

Northeastern University — Sep 2026 – May 2028
M.S. Computer Science, Silicon Valley campus — *San Jose, CA*

University of Wisconsin–Madison — May 2026
B.S. Data Science, Minor in Computer Science — *Madison, WI*

- **Coursework**: Big Data Systems, Artificial Intelligence, Machine Learning, Machine Organization, Programming III (Data Structures), Linear Algebra, Discrete Math, Data Science Modeling.

Technical Skills

Languages: Python, Rust, Go, C++, CUDA, TypeScript/JavaScript, SQL, R, Java
ML & Inference: PyTorch, vLLM, SGLang, Triton, Hugging Face, XGBoost, scikit-learn, RAG, conformal prediction, NumPy/Pandas
Systems & Infra: CUDA/PyO3 FFI, gRPC, Raft, LSM-trees, Kafka, AWS (Bedrock/S3), Docker, PostgreSQL, Redis, CI/CD, Linux
Web & Data: Next.js, React, Node, WebSockets, DeckGL/MapLibre, Drizzle, Flask, DuckDB

Experience

thaw (open-source LLM-inference infrastructure) — Apr 2026 – Present
Founder & Lead Engineer — *Rust · CUDA · Python*

- Created **thaw** ("git for live LLM agent sessions"): checkpoints, branches, diffs, and restores live vLLM/SGLang inference state (weights, KV cache, prefix-hash table, scheduler) so a session forks in **0.88s** median vs. ~340s cold boot on H100 (~400× amortized); **16 releases** shipped to PyPI as `thaw-vllm`, all benchmarks committed to the repo.
- Hit **14.3 GB/s** disk→GPU weight restore (3.4× faster 70B load) with a double-buffered `O_DIRECT` DMA pipeline overlapping disk reads, PCIe transfer, and an 8-shard parallel CRC32C verifier proven equal to a serial pass — bit-identical output across 8 models; hot-swaps an 8B model in **0.29s at 55 GB/s** (86% of PCIe Gen5 line rate) via a persisted pinned mmap.
- Removed a **60**× KV-cache snapshot bottleneck by coalescing ~16K per-block DMAs into one contiguous gather, and rebuilt vLLM's prefix-cache hash table on restore so forked sessions skip prefill entirely.
- Active in vLLM **RFC #34303** (CUDA checkpoint/restore): opened **PR #44074**, a pluggable sleep-mode backend abstraction, with receipted 70B sleep/wake integration on 2×H100; sole-authored *Re-feeding Is Not Replaying* (**arXiv:2606.15621**), measuring replay noise in counterfactual token-credit estimation (June 2026).
- Designed a `CudaBackend` trait (mock + real CUDA behind one contract) so the GPU pipeline is developed and tested without a GPU: **388 tests in CI** (155 Rust + 233 Python) on macOS and Linux with no CUDA toolchain.

Matteson Systems LLC — May 2026 – Present
Founder & Engineer — *Next.js · Postgres · Playwright · Claude*

- Shipped an **autonomous** SMB outreach product end to end: scraped and scored **10,500+** local businesses (OpenStreetMap), surfacing 158 high-priority leads in a single run at **~$0.03/lead**, costs receipted by a per-call ledger in integer microcents.
- Built a two-stage Claude-vision audit pipeline — real-device Playwright screenshots (Chromium + WebKit/iPhone), live Lighthouse Core Web Vitals, and on-page analysis — producing plain-English per-lead website report cards and personalized cold emails.
- Designed an append-only, event-sourced Postgres CRM (Drizzle migrations) and a threaded Gmail drip on Vercel cron (RFC822 in-thread replies, reputation-aware send caps) with built-in checkout and payments.

Research Cyberinfrastructure, UW–Madison DoIT — Jan 2026 – Apr 2026
AI Workflows Research Assistant — *AWS Bedrock · RAG · LLM eval*

- Benchmarked **9 LLMs** and 3 ensemble strategies across 282 questions on AWS Bedrock, building the team's evaluation and cost-tracking framework; ensemble majority voting (0.840) outperformed every individual model.
- Showed via Pareto analysis that Llama-4 Maverick reaches **98%** of top accuracy at a fraction of cost and latency, informing production model selection; presented at the UW–Madison ML+X Forum.

Selected Projects

Sentinel: Distributed Message Queue | github.com/matteso1/sentinel — *Go*

- Built a Kafka-inspired distributed log engine from scratch: custom **LSM-tree** storage (skip-list memtables at 3.9M reads/s, SSTables w/ CRC32, WAL, leveled compaction), a gRPC wire protocol, and topic/partition consumer groups.
- Implemented **Raft consensus** from scratch (leader election, log replication, split-brain prevention), validated by a deterministic in-memory network simulator covering partition + heal; 45 passing tests.

Madison Metro ML: Real-Time Bus Arrival Prediction | madisonbuseta.com — *Python, XGBoost, React*

- Shipped a live ML system correcting transit-API ETAs with a **47-feature** XGBoost model and **Mondrian conformal prediction** (calibrated 90%-coverage intervals stratified by route × day-type × horizon).
- Automated nightly retraining (GitHub Actions) behind a hard deployment gate ($\geq$2s MAE gain, no metric regression); React/DeckGL/MapLibre map rendering 200+ live vehicles at 60fps.

Also: Lattice (Rust+PyO3 reactive framework, Cranelift JIT, 10.6× vs. Streamlit) · brain2text (BCI Kaggle: 5-layer GRU + CTC decoder, 40% WER) · gitstare (Rust TUI, on crates.io) · LockBox (AES-256-GCM password manager).